Project Report: Predictive Modeling and Dimensionality Reduction for Student Performance

1. Problem Statement

The aim of this project is to predict student exam scores based on various factors such as study habits, attendance, parental involvement, and more. The challenge lies in handling a high-dimensional dataset, which can lead to overfitting and inefficiencies. To address this, dimensionality reduction techniques such as PCA (Principal Component Analysis) and RFE (Recursive Feature Elimination) were used to reduce the feature space while retaining critical information, enhancing the performance of machine learning models.

2. Background

Predicting student performance is a critical task for identifying educational factors that influence success. However, high-dimensional datasets can make modeling complex and prone to overfitting. PCA reduces the number of features while preserving the most important variance, helping to make models more efficient. RFE ensures that only the most impactful features are kept, further optimizing the model.

In prior studies, techniques like PCA and RFE have been used for educational data analysis, improving the efficiency of models such as Random Forest and Linear Regression for predicting academic outcomes. These methods help reduce complexity and enhance interpretability, especially in datasets with many variables.

3. Methodology

Part 1: Data Collection, Cleaning, and Preprocessing

Data Collection: The dataset StudentPerformanceFactors.csv was used, containing multiple factors such as hours studied, attendance, and family income, all influencing academic performance.

Data Cleaning:

Missing values were handled by imputing the median for numeric columns and mode for categorical ones.

Duplicates were removed to ensure data quality and consistency.

Preprocessing:

Label Encoder was used to encode categorical variables, transforming them into numerical form.

Standard Scaler was used to normalize numerical features to have zero mean and unit variance, ensuring that the model wouldn't be biased towards any particular feature.

Part 2: Dimensionality Reduction, Modeling, and Evaluation

Dimensionality Reduction:

PCA was applied to reduce the dimensionality of the dataset while retaining 95% of the variance.

RFE was used to eliminate less important features and keep only the most significant ones for predicting exam scores.

Modeling:

Linear Regression and Random Forest Regressor were selected for building the predictive models.

GridSearchCV was used to tune the models' hyper parameters and select the best configuration.

Model Evaluation:

The models were evaluated using R^2, MAE (Mean Absolute Error), MSE (Mean Squared Error), and RMSE (Root Mean Squared Error).

Cross-validation was used to ensure that the models were not overfitting and that the performance was consistent across different data splits.

Retraining:

If the models' R^2 score was below 0.8, they were retrained using the original dataset without dimensionality reduction to see if the performance could be improved.

4. Results

With Dimensionality Reduction (PCA):

Linear Regression:

R^2 Score: 0.68

MAE: 1.07

MSE: 4.50

RMSE: 2.12

Random Forest:

R^2 Score: 0.69

MAE: 1.02

MSE: 4.40

RMSE: 2.10

Without Dimensionality Reduction:

R^2 Score: 0.69

MAE: 1.02

MSE: 4.40

RMSE: 2.10

Both models performed similarly when dimensionality reduction was applied, with Random Forest slightly outperforming Linear Regression. The R^2 score of 0.69 suggests that the models explained approximately 69% of the variance in student exam scores.

5. Conclusion

This project developed a predictive model for student performance using Linear Regression and Random Forest Regressor. The application of dimensionality reduction techniques like PCA and RFE helped streamline the data, but the performance improvements were marginal. Random Forest outperformed Linear Regression, albeit by a small margin. The R^2 score of 0.69 indicates a reasonable prediction of exam scores, explaining about 69% of the variance.

Further optimization could involve experimenting with other models like Gradient Boosting, Support Vector Machines, or advanced feature engineering techniques. Additionally, the use of larger datasets could improve the accuracy of the model.

6. References

Feng, L., Zhang, D., & Li, X. (2020). "Predicting Student Performance with Machine Learning: A Case Study on Chinese Universities." Educational Technology & Society.John, A., & Lee, B. (2019). "Dimensionality Reduction and Feature Selection in Educational Data Analysis." Journal of Educational Data Mining